

Lab Goal : This lab was designed to teach you more about a linked list.

Lab Description : Write a single-linked linked list class. Each node in `SinglyLinkedList` has a data component and a reference to the next node only. Write and test all methods in `SinglyLinkedList`. Complete methods to add, remove, get, and search.

ListNode – stores a value and a reference to the next node

```
public class ListNode implements Linkable
{
    private Comparable listNodeValue;
    private ListNode nextListNode;

    public ListNode(){
        listNodeValue = null;
        nextListNode = null;
    }

    public ListNode(Comparable value, ListNode next){
        listNodeValue=value;
        nextListNode=next;
    }

    public Comparable getValue(){
        return listNodeValue;
    }

    public ListNode getNext(){
        return nextListNode;
    }

    public void setValue(Comparable value){
        listNodeValue = value;
    }

    public void setNext(Linkable next){
        nextListNode = (ListNode)next;
    }
}
```

Sample Data :

```
4 5 6 7 8 9 10 11 12 13
24 75 86 37 82 94 111 82 43
0 4 5 2 1 4 6
```

Sample Output :

Test add, addFirst, addLast, and toString:

```
LIST:: 4 5 6 7 8 9 10 11 12 13
LIST2:: 4 5 6 7 8 9 10 11 12 13
LIST3:: 20 4 5 6 7 8 9 10 11 12 13 40
```

Test get, getFirst, and getLast:

```
list.get(2) == 6
list.getFirst() == 4
list.getLast() == 4
```

Test size, contains, and equals:

```
list.size() == 10
list.contains(4) == true
list.contains(14) == false
list.equals(list)? == true
list.equals(list2)? == true
list.equals(list3)? == false
```

```

Test set, remove(int), remove(obj), removeFirst, and removeLast on list3:
LIST3:: 20 4 5 6 7 8 9 10 11 12 13 40
set(2, 100) returned 5
LIST3 after set(2, 100): 20 4 100 6 7 8 9 10 11 12 13 40
remove(2) returned 100
LIST3 after remove(2): 20 4 6 7 8 9 10 11 12 13 40
remove(4) returned true
LIST3 after remove(4): 20 6 7 8 9 10 11 12 13 40
remove(14) returned false
LIST3 after remove(14): 20 6 7 8 9 10 11 12 13 40
removeFirst() returned 20
LIST3 after removeFirst(): 6 7 8 9 10 11 12 13 40
removeLast() returned 40
LIST3 after removeLast(): 6 7 8 9 10 11 12 13

```

```

Test List Utility methods:
SUM:: 85.0
AVERAGE:: 8.5
SMALLEST:: 4
LARGEST:: 13

```

```

Test add, addFirst, addLast, and toString:
LIST:: 24 75 86 37 82 94 111 82 43
LIST2:: 24 75 86 37 82 94 111 82 43
LIST3:: 20 24 75 86 37 82 94 111 82 43 40

```

```

Test get, getFirst, and getLast:
list.get(2) == 86
list.getFirst() == 24
list.getLast() == 24

```

```

Test size, contains, and equals:
list.size() == 9
list.contains(37) == true
list.contains(100) == false
list.equals(list)? == true
list.equals(list2)? == true
list.equals(list3)? == false

```

```

Test set, remove(int), remove(obj), removeFirst, and removeLast on list3:
LIST3:: 20 24 75 86 37 82 94 111 82 43 40
set(2, 100) returned 75
LIST3 after set(2, 100): 20 24 100 86 37 82 94 111 82 43 40
remove(2) returned 100
LIST3 after remove(2): 20 24 86 37 82 94 111 82 43 40
remove(37) returned true
LIST3 after remove(37): 20 24 86 82 94 111 82 43 40
remove(100) returned false
LIST3 after remove(100): 20 24 86 82 94 111 82 43 40
removeFirst() returned 20
LIST3 after removeFirst(): 24 86 82 94 111 82 43 40
removeLast() returned 40
LIST3 after removeLast(): 24 86 82 94 111 82 43

```

```

Test List Utility methods:
SUM:: 634.0
AVERAGE:: 70.44444444444444
SMALLEST:: 24
LARGEST:: 111

```

```

Test add, addFirst, addLast, and toString:
LIST:: 0 4 5 2 1 4 6
LIST2:: 0 4 5 2 1 4 6
LIST3:: 20 0 4 5 2 1 4 6 40

```

```

Test get, getFirst, and getLast:
list.get(2) == 5
list.getFirst() == 0
list.getLast() == 0

```

```

Test size, contains, and equals:
list.size() == 7
list.contains(6) == true
list.contains(7) == false
list.equals(list)? == true
list.equals(list2)? == true
list.equals(list3)? == false

```

```
Test set, remove(int), remove(obj), removeFirst, and removeLast on list3:
LIST3:: 20 0 4 5 2 1 4 6 40
set(2, 100) returned 4
LIST3 after set(2, 100): 20 0 100 5 2 1 4 6 40
remove(2) returned 100
LIST3 after remove(2): 20 0 5 2 1 4 6 40
remove(6) returned true
LIST3 after remove(6): 20 0 5 2 1 4 40
remove(7) returned false
LIST3 after remove(7): 20 0 5 2 1 4 40
removeFirst() returned 20
LIST3 after removeFirst(): 0 5 2 1 4 40
removeLast() returned 40
LIST3 after removeLast(): 0 5 2 1 4
```

```
Test List Utility methods:
SUM:: 22.0
AVERAGE:: 3.142857142857143
SMALLEST:: 0
LARGEST:: 6
```

Empty List!